

Risk-Aware Sensor Placement Under Capability Uncertainty

Summary and Problem Formulation

The paper addresses a novel method for sensor deployment to detect ships passing through a predefined region of interest [1]. Crossings are represented by straight lines, which are transformed into points via a (α, p) transform in the representation space $\mathcal{C}$. Targets follow a log-Gaussian Cox line process, so in the representation space they form an inhomogeneous Poisson process with intensity λ estimated via integrated nested Laplace Approximation (INLA).

Detection model: Sensor i detects a trajectory l with probability (eq. 4 in [1]):

$$\gamma_i(l) = \rho_i \exp(-((x - a_x)^2 + (y - a_y)^2)/\sigma_l)$$

Where (x, y) is the point on the trajectory closest to the sensor

Let

$$g_i(l) = \exp(-((x - a_x)^2 + (y - a_y)^2)/\sigma_l)$$

We can write,

$$\gamma_i(l) = \rho_i g_i(l)$$

The paper assumes every sensor's maximum detection capability as known and fixed: ($\rho_i = \rho = 0.95$), and detection length scale $\sigma_l = 0.15$.

The void probability is defined by (eq. 2 in [1]):

$$P(N(\Theta) = 0) = E_\lambda[\exp(-\frac{1}{T_c} \int_\Theta \lambda(l) \pi_c(l, a) dl)]$$

Optimizing it directly is NP-hard, so the paper maximizes the void probability approximation (eq. 3 in [1]):

$$v(a) = \exp(-\frac{1}{T_c} \int_\Theta E_\lambda[\lambda(l)] \pi_c(l, a) dl)$$

Where $\lambda(l)$ is the target intensity, $\pi_c(l, a)$ the miss probability defined from the earlier $\gamma_i(l)$ by

$$\pi_c(l, a) = \prod_{i=1}^M (1 - \gamma_i(l))$$

Proposed Contribution: Introducing a Capability uncertainty

The paper assumes every sensor's maximum detection capability is known and fixed ($\rho_i = \rho = 0.95$).

In real life, sensors differ and so do their capability. I am proposing to model the sensor's capability as a belief, with each sensor's capability belief independent of the others:

$$\rho_i \sim N(\mu_i, \sigma_i^2), \quad i = 1, \dots, M$$

Where ρ_i is the random capability, μ_i the belief mean, and σ_i the standard deviation(uncertainty). A normal distribution is chosen to estimate ρ_i for convenience, and it allows for a closed form update (Gaussian conjugacy) [3].

With ρ_i random, the detection probability $\gamma_i(l) = \rho_i g_i(l)$ is random, hence so is the miss probability $\pi_c(l, a)$ and the void probability approximation $v(a, \rho)$. The implementation compares three planners:

- Nominal (baseline): No variance. The paper's setting
- Expected capability: $E_{\rho} [v(a, \rho)]$ – the mean of the distribution
- Risk-aware capability: maximizes $E_{\rho} [v(a, \rho)] - \kappa \text{Std}_{\rho} [v(a, \rho)]$

Design Choices / Assumptions

1. Capability belief is Gaussian. Tradeoff: Values can fall outside of $[0, 1]$. So we set the condition $\mu + 3\sigma < 1$ (three standard deviations above the mean)
2. Capabilities are independent across sensors, and of the intensity field.
3. Target intensity is fixed at its posterior mean. That's the paper's Eq. (3) approximation.
4. Sensors are static and placed once.

The next section addresses the mathematical implications of $\rho_i \sim N(\mu_i, \sigma_i^2)$

Mathematical analysis

Void Probability Approximation

We have the miss probability:

$$\pi_c(l, a) := \prod_{i=1}^M (1 - \gamma_i(l)) \text{ , with } \gamma_i(l) = \rho_i g_i(l)$$

By fixing sensor i and holding the others, we have:

$$\pi_c(l, a) := (1 - \rho_i g_i(l)) \prod_{j \neq i}^M (1 - \rho_j g_j(l)), \quad j \neq i$$

Which is affine in ρ_i . The expected missed crossings is therefore affine in ρ_i and can be written as:

$$U(\rho_i) = A - \rho_i B, \text{ with } A = \frac{1}{T_c \theta} \int \lambda \prod_j (1 - \rho_j g_j) dl, \quad B = \frac{1}{T_c \theta} \int \lambda g_i \prod_j (1 - \rho_j g_j) dl$$

Where A is the undetected mass with sensor i removed, and B is sensor i 's marginal coverage ($A, B > 0$).

Then:

$$v(\rho_i) = e^{-A} e^{\rho_i B}, \quad \frac{\partial v}{\partial \rho_i} = B e^{-A} e^{\rho_i B} \geq 0, \quad \frac{\partial^2 v}{\partial \rho_i^2} = B^2 e^{-A} e^{\rho_i B} \geq 0$$

v is convex in ρ_i .

Jensen Gap Analysis

Due to the convexity of v in ρ_i , we can write for a sensor i :

$$E_{\rho_i} [v] \geq v(E_{\rho_i} [\rho_i]) \Rightarrow E_{\rho_i} [v] \geq v(\mu_i)$$

Jensen on ρ_1 and taking expectation of $(\rho_2 \dots \rho_M)$ on each side:

$$E_{\rho_1} [v | \rho_2 \dots \rho_M] \geq v(\mu_1, \rho_2 \dots \rho_M)$$

After M steps, we have:

$$E_{\rho} [v] \geq v(\mu)$$

The paper analyzes the analogous gap for its random intensity field λ and shows it small [2]; the capability gap here is the same Jensen structure over ρ instead of λ .

The nominal planner uses μ for each sensor's capability. For a sensor i , The Gaussian moment-generating function gives:

$$E_{\rho_i}[v] = v(\mu_i)e^{\frac{1}{2}B^2\sigma_i^2}, \quad gap = v(\mu_i)e^{\frac{1}{2}B^2\sigma_i^2} - v(\mu_i) \approx \frac{1}{2}B^2\sigma_i^2v(\mu_i)$$

The gap grows with the variance σ_i^2 and squared sensor's marginal coverage B^2 . This means that if $B \approx 0$, there is no gap and a change in the uncertainty will produce the same result.

The expected detection coverage $D(a)$ can be written as:

$$D(a) = \frac{1}{T_c} \int_{\Theta} \lambda(l) \gamma_i(l) dl = \frac{1}{T_c} \int_{\Theta} \lambda(l) (1 - \pi_c(l, a)) dl$$

D is monotone submodular in the sensor set, so greedy selection attains the $(1 - 1/e)$ guarantee [2]. We can write:

$$E_{\rho}[D(a)] = \pi_c(\mu) \Rightarrow E_{\rho}[D(a)] = D(a; \mu)$$

Under uncertainty

Nominal and Expected select the same placement: expected coverage is linear in capability, so $E_{\rho}[D(a)] = D(a; \mu)$, and greedy on either achieves the same result. The implementation therefore compares Nominal against Risk-aware planners.

Simulation / Results

To demonstrate the results, the uncertainty scale factor is selected in the range $[0, 3]$. The empty-placement void probability $e^{-\Lambda} = 0.146$. Placing five sensors raises the void probability to 0.3284, which confirms that detection lowers the undetected mass.

Two planners are compared. Nominal maximizes v at the belief mean and is blind to σ . Risk-aware maximizes $E[v] - kStd[v]$, estimated by Monte Carlo, with $\kappa = 10$.

It is observed from the simulation results that:

1. Placement geometry is unchanged by uncertainty. Both planners place sensors on the same five locations across the sweep (Figure A).
2. The difference is in outcome spread, while the mean is identical. As σ grows, both placements yield the same mean void probability (0.3284), but different distributions (Figure B).
3. Risk-aware v dominates on the downside. The 5th-percentile void probability holds flat for the risk-aware planner and declines steadily for the nominal planner as uncertainty grows (Figure C).
4. The Jensen gap matches the analysis (Figure D). On a fixed placement, the realized gap $E[v] - v(\mu)$ grows linearly in σ^2 , tracking the predicted $\frac{1}{2}B^2\sigma^2v(\mu)$

Table 1: Planner comparison. Void probability statistics over the capability belief, per uncertainty scale.

sigma_scale	sigma_large	nominal mean	nominal q05	risk mean	risk q05	q05 gap
0.0	0.0000	0.32841	0.32824	0.32841	0.32841	+0.00017
0.5	0.0035	0.32841	0.32782	0.32841	0.32824	+0.00042
1.0	0.0070	0.32841	0.32723	0.32841	0.32824	+0.00101
1.5	0.0105	0.32841	0.32665	0.32841	0.32824	+0.00159
2.0	0.0140	0.32842	0.32606	0.32841	0.32824	+0.00218
3.0	0.0210	0.32843	0.32489	0.32841	0.32824	+0.00334

The means are equal across planners. Nominal q05 decreases with σ . The q05 gap widens from 0.00017 to 0.00334 across the sweep.

Table 2: Jensen-gap validation. Realized gap against the first-order prediction, on a single fixed placement with all five units uncertain.

sigma_square	realized gap	Std error mean	predicted	ratio
0.00e+00	0.00e+00	0.00e+00	0.00e+00	—
1.23e-05	2.57e-07	8.07e-07	1.99e-07	1.29
4.90e-05	9.10e-07	1.61e-06	7.96e-07	1.14
1.10e-04	1.96e-06	2.42e-06	1.79e-06	1.09
1.96e-04	3.41e-06	3.23e-06	3.19e-06	1.07
4.41e-04	7.49e-06	4.84e-06	7.17e-06	1.05

The ratio approaches 1 as σ grows. At small σ the realized gap runs slightly above the prediction. With several uncertain sensors, each one adds its own gap and the totals sum, a consequence of the sensors being independent.

Open Problems / Discussion

1. Correlated capabilities: The analysis assumes sensors are independent. This drives the per-coordinate Jensen bound and the additive gap in Table 2. Under correlation the cross terms no longer vanish, and both the bound and the gap need the full covariance.
2. Static placement: Sensors are placed once. Moving them to cut uncertainty where it matters most ties placement to the belief update, outside this one-time setup.
3. Risk measure. The objective penalizes $Std[v]$, which treats upside and downside spread the same.

References

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- [3] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006. Available: <https://link.springer.com/book/9780387310732>